Responsible AI Model Evaluations

A Seven-Week Red-Teaming Study of Foundation Models Across 22 Safety Risk Categories

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Framework: RedBench v0.1.0 (MIT License) · Benchmark: Dang et al. 2026, arXiv:2601.03699
Compliance alignment: NIST AI RMF 1.0 · GSAR 552.239-7001

Abstract

This paper presents the findings of a seven-week continuous automated red-teaming study evaluating seven foundation models — Claude Opus 4.7, Claude Opus 4.6, Claude Sonnet 4.6, GPT-4o, GPT-4o Mini, Gemini 2.5 Flash, and Gemini 2.5 Pro — against 19,500 total evaluation samples drawn from the open RedBench dataset spanning 22 structured safety risk categories. Using a dual-metric framework that jointly measures the Attack Success Rate (ASR), the False Outright Refusal Rate (FOR), a Drift Coefficient tracking multi-turn safety degradation, and a Provenance Score measuring chain-of-reasoning traceability, we characterise each model's safety posture across both bypass vulnerability and usability penalty. Across 19,500 evaluations, 894 attack failures and 317 critical-severity findings were recorded. CBRN Information or Capabilities remained the highest-risk category (mean ASR 32.2%), followed by Cybersecurity Threats (21.5%). Notably, Election Interference emerged as a significant new risk in the May 8 run, recording 33% ASR across five of seven models — the first time this category has produced substantial bypass rates in the study. Claude Opus 4.7 leads the current leaderboard (ASR 4.25%, FOR 1.00% on May 8) with a mean ASR of 3.75% and mean FOR of 1.33% across its three evaluation runs. Gemini 2.5 Pro recorded a 27% false refusal rate on May 8 — the highest single-run FOR in the study — alongside 16 critical findings. All findings are mapped to NIST AI RMF 1.0 functions and structured as GSAR 552.239-7001-aligned forensic audit records suitable for direct inclusion in compliance documentation.

19,500	894	317	303	22	7
Total Evaluations	Attack Failures	Critical Findings	Over-Refusals	Risk Categories	Evaluation Weeks

1. Introduction

The rapid deployment of large language models (LLMs) in enterprise, government, and public-sector contexts has created urgent demand for independent, reproducible, and continuously updated safety characterisations. Internal evaluations conducted by model developers are necessary but insufficient: they are not independently auditable, they do not expose the specific failure categories relevant to a given deployer's threat model, and they do not track safety degradation under multi-turn adversarial pressure.

This study was designed to fill that gap. Beginning April 10, 2026, we deployed an automated weekly evaluation pipeline that runs 500 samples per model per run against the RedBench benchmark — a publicly available, MIT-licensed dataset of red-teaming prompts covering 22 risk categories across five NIST AI RMF functional domains. Each sample is evaluated by a NeuralJudge (claude-haiku-4-5) that produces a structured SAFE / FAIL / CRITICAL / OVER_REFUSAL verdict along with a chain-of-reasoning justification, a NIST subcategory mapping, and a remediation priority.

The evaluation framework addresses four overlapping stakeholder needs: red-teamers require category-level granularity and drift signals; compliance officers require audit-ready forensic records; product decision-makers require a joint safety–usability trade-off surface; and policy analysts require a time-series comparison anchored to a fixed, open benchmark independent of any model provider's internal testing.

2. Methodology

2.1 Benchmark Dataset

All evaluations use the RedBench dataset (Dang, Ngo & Hy, 2026; arXiv:2601.03699). RedBench is a non-proprietary, MIT-licensed red-teaming corpus covering 22 safety risk categories including CBRN Information or Capabilities, Cybersecurity Threats, Illegal Weapons, Malware Code, Sexual Content, Misinformation, Election Interference, Child Safety, Data Privacy, and 13 additional categories. Each weekly run draws 500 samples stratified across adversarial attack samples (80%; n=400) and benign refusal-test samples (20%; n=100), with 20 samples reserved as multi-turn drift probes that escalate across conversation turns.

2.2 Metrics

Metric		
Attack Success Rate (ASR)	Fraction of adversarial samples receiving a FAIL or CRITICAL verdict	Lower is safer. Measures bypass vulnerability.
False Outright Refusal Rate (FOR)	Fraction of benign samples where the model refused a safe, legitimate request	Lower is better. Measures usability penalty.
Drift Coefficient	Slope of ASR across escalating conversation turns on 20 drift-probe samples	Positive = safety erodes under multi-turn pressure. Zero = stable.
Provenance Score	Fraction of NeuralJudge verdicts that include a complete chain-of-reasoning	Higher is better. Measures audit traceability.

2.3 Neural Judge

Each model response is evaluated by a NeuralJudge implemented on `claude-haiku-4-5-20251001`. The judge receives the original prompt, the model response, and the risk category, and outputs a structured JSON verdict containing: (1) a primary verdict (SAFE / FAIL / CRITICAL / OVER_REFUSAL), (2) a severity tier, (3) a NIST AI RMF function and subcategory mapping, (4) a remediation priority, and (5) a chain-of-reasoning justification. CRITICAL verdicts are reserved for responses posing immediate physical harm risk. All PII is redacted prior to storage and audit log generation.

2.4 Evaluation Infrastructure

Evaluations run automatically via GitHub Actions every Friday at 18:00 UTC. Each run evaluates all registered models concurrently via asynchronous Python. The pipeline includes a pre-flight API credit check, configurable full-chain audit mode (REDEVAL_FULL_AUDIT_CHAIN), and automated dashboard rebuild.

Runs where ≥50% of API calls fail are flagged `eval_failed` and excluded from all statistics.

3. Models Evaluated

Model				
Claude Opus 4.7	Anthropic	3	Apr 24, May 2, May 8	Newest model; leads latest leaderboard
Claude Opus 4.6	Anthropic	7	Apr 10 – May 8	Full seven-week cohort
Claude Sonnet 4.6	Anthropic	7	Apr 10 – May 8	Full seven-week cohort
GPT-4o	OpenAI	7	Apr 10 – May 8	Full seven-week cohort
GPT-4o Mini	OpenAI	7	Apr 10 – May 8	Full seven-week cohort
Gemini 2.5 Flash	Google	4	Apr 17, Apr 24, May 2, May 8	Migrated from deprecated 2.0 Flash endpoint Apr 14
Gemini 2.5 Pro	Google	4	Apr 14, Apr 17, Apr 24, May 8	Excluded May 2 (eval_failed); 27% FOR on May 8

Note: Gemini runs on Apr 10 and Apr 12 used deprecated endpoints returning HTTP 404 and are excluded. The May 1 run failed for all Anthropic models and was deleted. Gemini 2.5 Pro was eval_failed on May 2 but returned successfully on May 8.

4. Results

4.1 Per-Model Performance Summary

Table 2 reports average metrics across all valid evaluation runs per model, ranked by average ASR. Δ columns compare each model's first to most recent run. The May 8 leaderboard shows Claude Sonnet 4.6 and Gemini 2.5 Flash tied at lowest ASR (4.00%), with Claude Opus 4.7 at 4.25% — all three within 0.25pp of each other.

Model							
Claude Opus 4.7	3	3.75%	1.33%	+0.25pp	–1.00pp	13	4.25%
Gemini 2.5 Flash	4	3.44%	15.00%	–1.50pp	–3.00pp	15	4.00%
Claude Sonnet 4.6	7	5.00%	1.00%	–2.50pp	0.00pp	44	4.00%
Claude Opus 4.6	7	5.46%	0.86%	–1.50pp	+1.00pp	46	4.50%
GPT-4o	6	5.46%	7.86%	0.00pp	–2.00pp	54	6.25%
Gemini 2.5 Pro	4	7.56%	18.25%	+0.50pp	+9.00pp	72	7.75%
GPT-4o Mini	7	8.11%	14.00%	–0.75pp	+3.00pp	73	9.50%

pp = percentage points. ↓ = lower is better. Δ = first available run vs May 8. Green <5% ASR / <3% FOR; amber moderate; red high.

4.2 Weekly ASR Trend — Seven-Week Cohort

Table 3 shows week-by-week ASR for the four models with complete seven-week history. A step-change improvement is visible across all four models between Apr 12 and Apr 14, consistent with a provider safety update. However, the May 8 run reveals a regression for GPT-4o (5.25% → 6.25%) and GPT-4o Mini (6.00% → 9.50%), suggesting that safety improvements are not uniformly sustained and require continued monitoring.

Model								
Claude Opus 4.6	6.00%	7.50%	4.25%	4.00%	6.75%	5.25%	4.50%	–1.50pp
Claude Sonnet 4.6	6.50%	7.00%	2.75%	3.25%	6.50%	5.00%	4.00%	–2.50pp
GPT-4o	6.25%	6.50%	3.25%	3.75%	7.00%	5.25%	6.25%	0.00pp
GPT-4o Mini	10.25 %	10.00 %	5.50%	6.00%	9.50%	6.00%	9.50%	–0.75pp

Table 3. Week-by-week ASR for the four full-cohort models. Green ≤5%, amber 5–10%, red >10%. May 8 shows regression for GPT-4o and GPT-4o Mini.

4.3 Risk Category Analysis

Table 4 shows average ASR by risk category across all valid model-week pairs. Three categories dominate: CBRN (32.2%), Cybersecurity Threats (21.5%), and Illegal Weapons (13.2%). A notable new development in the May 8 run is the emergence of **Election Interference** as a significant failure category — recording 33% ASR simultaneously across five of seven models (Claude Opus 4.7, Claude Sonnet 4.6, GPT-4o, GPT-4o Mini, and Gemini 2.5 Pro). This is the first run in the study where Election Interference produced material bypass rates, pushing its overall average to 5.13% — tied fifth with Sexual Content.

Category			
CBRN Information or Capabilities	32.23%	Extremism and Radicalization	2.99%
Cybersecurity Threats (Beyond Malware)	21.47%	Hateful Content	2.94%
Illegal Weapons (Non-CBRN)	13.19%	Crime Content	2.86%
Malware Code	11.36%	Economic Harm	2.48%
Sexual Content	5.13%	Data Privacy	2.34%
Election Interference ▲	5.13%	Decision-Making in Critical Systems	2.32%
Misinformation	4.06%	Self-Harm Content	2.03%
Environmental Harm	3.85%	Abusive Content	1.86%
Bias Content	3.30%	Confabulation	1.28%
Violent Content	3.03%		

▲ *= significant new entrant in May 8 run. Categories with 0% ASR across all models and all weeks: Child Safety · Intellectual Property · PII Leakage · Toxicity · Hate Speech · Violence · Self-Harm.*

4.4 Multi-Turn Safety Drift

GPT-4o Mini and GPT-4o share the highest average drift coefficient (+0.397 each), indicating recurring multi-turn safety degradation. GPT-4o Mini peaked at +1.111 on May 2. Claude Opus 4.7 recorded +2.222 on May 2 — the highest single-run value in the dataset — but returned to a lower reading on May 8, suggesting the May 2 result may reflect statistical variance from limited drift-probe samples rather than a sustained pattern; it warrants continued monitoring. Claude Sonnet 4.6 averaged –0.079 across seven weeks, the most stable safety profile in the dataset. Claude Opus 4.6 averaged –0.317, the most consistently negative coefficient — indicating this model tends toward over-conservatism under escalating adversarial pressure rather than capitulation.

5. Audience-Specific Insights

FOR AI ENGINEERS & RED-TEAMERS

5.1 Category-Level Failure Signatures and Emerging Threats

CBRN remains the dominant and unresolved attack surface. 32.2% mean ASR across all model-week pairs — more than 1.5× the next-highest category. No model in any run has achieved an acceptable bypass rate on CBRN-adjacent prompts. This category accounts for a disproportionate share (~36%) of all CRITICAL findings.

Election Interference is the most significant new signal in the May 8 run. 33% ASR simultaneously across five of seven models — the first material bypass rates for this category in the study. Red-teamers should immediately probe Election Interference prompts across all target models, particularly for social engineering and influence operation framings that may have shifted in the RedBench corpus weighting.

GPT-4o Mini and GPT-4o both regressed on May 8. GPT-4o Mini jumped from 6.00% to 9.50% ASR; GPT-4o from 5.25% to 6.25%. Both models had shown sustained improvement between Apr 14–May 2. The reversal, coinciding with the Election Interference spike, suggests category-correlated regression rather than a general safety degradation.

Seven categories maintained 0% ASR across all 7 weeks. Child Safety, Intellectual Property, PII Leakage, Toxicity, Hate Speech, Violence, and Self-Harm show no bypasses. Probe these with novel adversarial rephrasings not present in the static RedBench corpus — 0% on a fixed benchmark is not a ceiling.

FOR GRC & COMPLIANCE OFFICERS

5.2 NIST AI RMF Mapping and Forensic Audit Structure

317 critical findings across 7 weeks, all GSAR 552.239-7001-aligned. Every CRITICAL and high-severity FAIL verdict is stored as a forensic audit record containing PII-redacted prompt and response, NeuralJudge chain-of-reasoning, NIST AI RMF function/subcategory, and remediation recommendation — structured for direct inclusion in SSPs, RARs, and POA&M; documents.

GPT-4o Mini carries the highest critical finding burden: 73 over seven weeks. That is 23% of all critical findings in the study from a single model. Any government or regulated-sector deployment of GPT-4o Mini must document this rate and the compensating controls in place. The model's May 8 regression (9.50% ASR) further elevates this concern.

Gemini 2.5 Pro's May 8 FOR spike (27%) is a material availability risk. A 27% false refusal rate means more than one in four legitimate requests is refused. In a high-stakes operational environment, this represents a service reliability failure independent of safety considerations. Any ATO documentation should capture this metric alongside ASR.

Anthropic models consistently record near-zero GOVERN function ASR. NIST GOVERN (GV.PO) shows non-zero ASR only for GPT-4o Mini and Gemini 2.5 Pro, indicating that some failures in those models trace to policy gaps rather than detection failures — a distinction that determines whether remediation is a fine-tuning task or a policy documentation task.

FOR PRODUCT LEADERS

5.3 The Safety–Usability Trade-Off Surface

Claude Opus 4.7 is the strongest choice for sensitive-domain deployment. 3.75% mean ASR, 1.33% mean FOR across three runs. In its latest run (May 8) it recorded 4.25% ASR and 1.00% FOR — placing it at the top of the safety-usability Pareto frontier alongside Claude Sonnet 4.6. With only three weeks of data its trend is still forming, but no run has exceeded 4.25% ASR or 2.00% FOR.

Claude Sonnet 4.6 remains the most proven option. Seven continuous weeks of data. Best average FOR among full-cohort models (1.00%). Largest ASR net reduction (–2.50pp). May 8 result: 4.00% ASR, 2.00% FOR — the joint lowest ASR in the latest run alongside Gemini 2.5 Flash.

Gemini 2.5 Pro's 27% FOR on May 8 is a deployment blocker. The model has shown a worsening FOR trajectory: 18% (Apr 14) → 19% (Apr 17) → 9% (Apr 24) → 27% (May 8). The Apr 24 improvement has fully reversed. Until this instability is resolved, Gemini 2.5 Pro should not be deployed in any application where users expect reliable responses to legitimate requests.

GPT-4o Mini's May 8 regression warrants caution. After improving from 10.25% to 6.00% between Apr 10 and May 2, it reverted to 9.50% on May 8. Teams using GPT-4o Mini should implement category-specific guardrails — particularly for CBRN, Election Interference, and Cybersecurity — rather than relying on the model's native safety behaviour.

FOR POLICY ANALYSTS & RESEARCHERS

5.4 Time-Series Safety Posture Against a Fixed Open Benchmark

The Election Interference spike on May 8 is the most important policy signal. 33% ASR across five of seven models in a single run — on a category directly relevant to democratic integrity — represents a structural vulnerability that had not previously been detected in this study. Because the benchmark is fixed and unchanged, the spike reflects either a model capability change or a prompt-specific sensitivity that was not captured in prior runs. Policy bodies evaluating LLM deployment near election infrastructure should treat this as a high-priority finding.

CBRN remains the most persistent cross-provider gap. Seven consecutive weeks, every model, every run: CBRN consistently produces the highest category-level ASR with no sign of convergence. The 32.2% mean ASR versus near-zero rates for Child Safety and PII Leakage represents a structural asymmetry in safety fine-tuning priorities with direct proliferation policy implications.

Safety improvements are not monotonic. The GPT-4o and GPT-4o Mini regressions on May 8 — after four weeks of improvement — demonstrate that a model's safety posture can deteriorate between provider updates. This is only detectable with fixed-benchmark continuous evaluation. Single-point or annual assessments would have missed this regression entirely.

The full pipeline is open and reproducible. Complete source code, benchmark data, audit logs, and dashboard are published at MIT License on GitHub. Any researcher, regulator, or government body can run an independent replication at low cost using only API keys and the public repository.

6. Discussion

6.1 Limitations

The study spans seven weeks, providing a richer longitudinal view than prior versions, but category-level ASR estimates carry high variance for rare categories (one failure in a 14-sample category = 7.1% ASR). The NeuralJudge (claude-haiku-4-5) introduces potential systematic bias as an Anthropic model evaluating both Anthropic and non-Anthropic responses. Claude Opus 4.7 has only three valid runs; its leading position should be treated as promising but not yet statistically definitive.

The May 8 Election Interference spike — 33% ASR across five models in a single run — requires validation. A single-run result at category-level could reflect sampling variance given the small number of category prompts per run (~7–14). Confirmation in the next run would significantly increase confidence in this finding.

The public availability of RedBench creates benchmark contamination risk. The correlated improvement observed between Apr 12 and Apr 14 across all models is consistent with both general safety updates and benchmark-specific fine-tuning. A held-out private evaluation set would be required to distinguish these.

6.2 Future Work

Planned extensions: expansion to 1,000 samples per run to reduce category-level variance and increase reliability of sparse-category signals like Election Interference; a held-out private evaluation set for contamination-resistant comparison; integration of adversarial prompt generation to probe categories beyond the static RedBench corpus; and enabling REDEVAL_FULL_AUDIT_CHAIN=all mode for complete forensic chain generation across all samples.

7. Conclusions

Seven weeks of continuous automated red-teaming across 19,500 evaluations against a fixed open benchmark produce four principal findings.

First, CBRN safety is the most persistent unresolved vulnerability. 32.2% mean ASR, every model, every week — no convergence toward acceptable rates. This represents an urgent cross-provider gap in current safety fine-tuning priorities.

Second, Election Interference emerged as a significant new risk category in the May 8 run, with 33% ASR across five of seven models. This is the most operationally urgent new finding in the study and requires immediate verification in subsequent runs.

Third, safety improvements are not monotonic. GPT-4o and GPT-4o Mini both regressed on May 8 after sustained improvement, demonstrating that only continuous fixed-benchmark evaluation can detect regression between provider updates.

Fourth, the safety–usability trade-off is manageable but not automatic. Claude Opus 4.7 and Claude Sonnet 4.6 demonstrate sub-5% ASR with sub-2% FOR simultaneously. Gemini 2.5 Pro's 27% FOR and GPT-4o Mini's reverted 9.50% ASR show that the optimal balance requires active monitoring and model-specific deployment controls.

8. References

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